SOC Internship Program – Day 25 Notes

User and Entity Behavior Analytics (UEBA)

1. Introduction to UEBA

UEBA stands for **User and Entity Behavior Analytics**.
It is a cybersecurity technique used in Security Operations Centers (SOC) to analyze the behavior of users and systems and detect suspicious or abnormal activities.

UEBA uses:

- Machine Learning
- Behavior Analysis
- Anomaly Detection

to identify threats that traditional security tools may miss.

2. Objectives of UEBA

The main objective of UEBA is to:

- Detect insider threats
- Identify compromised accounts
- Monitor unusual activities
- Improve threat visibility
- Reduce false positives

UEBA helps SOC analysts understand what is "normal" behavior and quickly detect deviations from that behavior.

3. Why UEBA is Important

Traditional security systems mainly depend on predefined rules.
Modern attackers often bypass these rules using stolen credentials or insider access.

UEBA is important because it can:

- Detect attacks without predefined signatures
- Identify suspicious user activities
- Detect account compromise
- Improve security monitoring efficiency

Examples:

- Employee logging in at unusual times
- Sudden large file downloads

- Access from unknown locations

4. Common UEBA Indicators

UEBA tools monitor several suspicious indicators such as:

Login-Based Indicators

- Login from unusual country
- Impossible travel activity
- Multiple failed login attempts
- Login outside office hours

Data Access Indicators

- Large data downloads
- Access to confidential files
- Access to unusual systems

User Behavior Indicators

- Sudden privilege escalation
- Unusual application usage
- Abnormal network activity

5. UEBA Detection Logic

UEBA works by comparing:

- Normal behavior
  with
- Current activity

Example:

Normal Behavior

- User logs in daily from office location during working hours.

Abnormal Behavior

- User logs in at 3 AM from another country.

Result

- UEBA system generates an anomaly alert.

6. Example UEBA Investigation

Alert Generated

User downloaded **20GB sensitive data**.

Findings

- Login from unusual IP address
- Activity outside working hours
- Access to sensitive resources

Analysis

Possible:

- Insider Threat
  or
- Compromised Account

SOC analysts investigate whether the activity is legitimate or malicious.

7. UEBA vs Traditional SIEM

Feature	Traditional SIEM	UEBA
Detection Type	Rule-Based	Behavior-Based
Detection Method	Signature & Rules	Machine Learning
Focus	Events & Logs	User Behavior
Insider Threat Detection	Limited	Strong
Unknown Threat Detection	Difficult	Better Detection

8. SOC Workflow Using UEBA

Workflow Steps

1. User Activity Logs Collected
2. Behavior Analysis Performed
3. Anomaly Detection Triggered
4. SOC Investigation Started

9. SOC Team Responsibilities

L1 Analyst

- Monitor anomaly alerts
- Identify suspicious activities

L2 Analyst

- Investigate abnormal behavior
- Correlate logs and alerts

L3 Analyst

- Tune UEBA models
- Improve detection logic

- Reduce false positives

Class Activity Analysis

Scenario

- Employee logs in from India
- 10 minutes later logs in from Germany
- Downloads confidential files

Why is this Suspicious?

This is suspicious because:

- Impossible travel detected
- Unusual login location
- Sensitive data download activity
- Possible stolen credentials
- Potential insider threat

Assignment Answers

1. Explain UEBA

UEBA is a cybersecurity technology that analyzes user and entity behavior to detect anomalies and suspicious activities using behavior analytics and machine learning.

2. Differentiate SIEM and UEBA

SIEM	UEBA
Rule-based detection	Behavior-based detection
Detects known attacks	Detects unknown threats
Focus on logs	Focus on behavior
Limited insider detection	Strong insider threat detection

3. Analyze Insider Threat Scenario

If an employee accesses sensitive files outside office hours from an unusual location and downloads large amounts of data, it may indicate malicious insider activity or account compromise.

4. List Abnormal Behavior Indicators

- Unusual login location
- Abnormal login time
- Large file downloads

- Multiple failed logins
- Access to unusual systems
- Impossible travel activity